Quantum mesh partitioning for heterogeneous devices: unequal work shares expose a structural limit of a varQITE ansatz

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Abstract. Quantum graph partitioning has so far been studied for *balanced* bipartitions, notably by IonQ with Ansys and Synopsys inside LS-DYNA. Modern clusters mix CPUs and GPUs, so a mesh must be split with an *unequal* target share α proportional to device speed. We formulate this heterogeneous split as an Ising problem and benchmark QAOA, its variants and IonQ's variational imaginary-time evolution (varQITE) against classical heuristics on nine reproducible finite-element meshes, grading every method against the exact optimum and counting every evaluation. The heterogeneous case changes the picture: varQITE with the published pair-rotation ansatz returns a near-optimal split on 78% of shots for equal devices but fails completely for α=0.8. We prove why: a qubit that only ever acts as the control of the ansatz's two-qubit gates keeps a 50/50 marginal for all angles, which caps the probability of a unique optimum at 1/2; equal shares hide this because their optimum is doubly degenerate. QAOA beats random sampling at both shares and matches local search at 12 qubits, but falls behind classical heuristics from 16 qubits on. At 29 qubits (536 870 912 splits, simulated on HPC and graded against the exact optimum), the best QAOA settings found at 20 qubits still beat uniform guessing per shot by 1 204× / 2 279× (α=0.5 / 0.8). At α=0.8, however, they reach only 0.7× the per-shot success of a biased coin that sends each region to the GPU with probability α. 1 000 shots contain a near-optimal split with probability 0.74% / 0.7%, against 10% / 43% for simulated annealing with the same budget and 50% / 100% for spectral bisection. No quantum advantage is claimed.

1 Introduction and contributions

Parallel PDE solvers partition their mesh among compute devices; the partition sets both the load balance and the communication volume. IonQ inserted quantum graph partitioning into Ansys/Synopsys LS-DYNA to reduce fill-in of sparse direct solvers, first with varQITE [1, 2] and later with Iterative-QAOA [3]; quantum annealing has been tried for HPC load balancing [4]. These studies use balanced halves and closed code. Our contributions:

1. **Heterogeneous partitioning**: a two-device split with target share $\alpha = s_B/(s_A + s_B)$ from device speeds, plus a device-choice layer for a synthetic multi-device pool.
2. **A structural limit of the varQITE ansatz under unequal shares** (Lemma 1), measured on all cases, and a one-layer change (A^+) that removes the cap.
3. **An open benchmark graded against the exact optimum**: nine meshes, five QAOA families, varQITE and five classical baselines, with total evaluations counted for every method and all tuning done on held-out meshes; the best settings are re-tested at 29 qubits on HPC, still against the exact optimum.
4. **Reproducibility**: meshes and region assignments shipped as data files; 30 injected bugs all caught by the checks; one batched IonQ job prepared.

2 Problem and model

Benchmark. Nine plates meshed by Delaunay triangulation (1 670–2 166 triangles, Fig. 1; regions in Fig. 2): a rectangle with a known answer, an L-plate with a heat source, two sources, a plate with a hole (stress-style mesh), and a five-step complexity ladder on one L-plate (uniform → three sources → heavy-material zone, 3× work → strong-coupling zone, 3× border cost → priority zone). Each mesh is grouped into n connected regions (one qubit each) by k-means with a connectivity repair; region work a_i and border cost c_{ij} are exact sums. Meshes and region assignments are stored as data files, because floating-point differences made 4 of 9 meshes differ between macOS/arm64 and Linux/x86-64 builds.

Ising model. With z_i=+1 (device A) or −1 (device B) and normalised $\tilde{a}_i$, $\tilde{c}_{ij}$,

$$E(z) = \sum_{\langle ij\rangle} \tilde{c}_{ij}\tfrac{1-z_iz_j}{2} + \lambda\Big(\sum_i \tilde{a}_i z_i - (1-2\alpha)\Big)^2 \; [+\mu P(z)],$$

the Ising form of graph partitioning [5] with a target share. For $\alpha\neq0.5$ the balance term contains single-qubit fields $h_i \propto (1-2\alpha)\tilde{a}_i$, so $E(z) \neq E(-z)$ and the optimum is generically unique. The balance term couples every pair of regions ($n(n-1)/2$ two-qubit terms), which suits IonQ's all-to-all connectivity. Scenarios: equal devices (α=0.5, λ=10) and CPU+GPU with an assumed 4× faster GPU (α=0.8, λ=3.98); λ is shared per scenario and validated by exact enumeration.

Device choice (modelled). For a synthetic pool of 8 devices with assumed speed, memory and power, exhaustive search gives a fastest plan of 9.4 h / 27.3 kWh, an overnight plan (≤12 h) of 11.8 h / 23.7 kWh and a lowest-energy plan of 13.4 h / 22.0 kWh, against 37.5 h / 56.1 kWh for equal shares.

3 Methods

QAOA (noiseless statevector, cross-checked against Qiskit circuits to 10^{-15}): cost layer $e^{-i\gamma E}$, mixer $e^{+i\beta\sum X}$; linear-ramp LR-QAOA [6]; optimised angles with COBYLA, Nelder–Mead, Powell, L-BFGS-B (budget 200 evaluations) and shot-based SPSA [7]; CVaR-QAOA [8]; warm starts [9]; a symmetry-reduced variant at α=0.5. **varQITE** [2]: each step solves $G\dot{\theta} = d$ by least squares with $d_k = -(\langle P_kH\rangle - \langle P_k\rangle\langle H\rangle)$ and $G_{kj} = \partial\langle P_k\rangle/\partial\theta_j$ (parameter shift), starting from $|+\rangle^{\otimes n}$ with all angles zero. Ansätze: the paper's $e^{i\theta Z_rY_q}$ on all pairs ordered by vertex weight (ZY); R_Y on the target followed by CX on all pairs or on graph edges (A); A with one R_Y per qubit in front (A^+, ours); an IonQ-native RZZ + RX ansatz (B). Training uses exact expectations. **Classical**: uniform and target-share random sampling, simulated annealing, local search with restarts, spectral bisection. **Metrics**: the probability per shot of a split with cost ≤ 1.05× optimum, and the probability that 1 000 shots or candidates contain one. **Total evalua-**

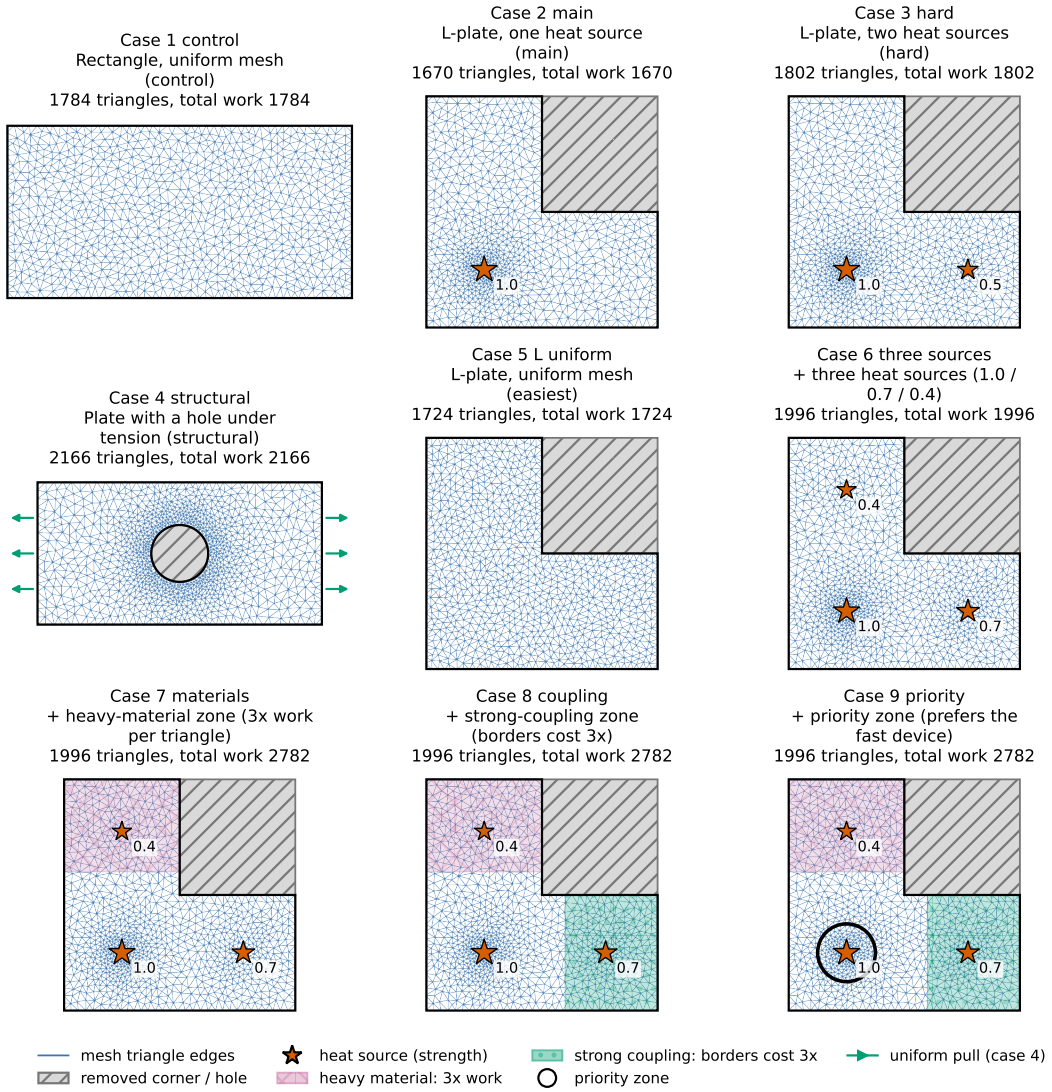


Figure 1: The nine benchmark objects (seed-7 meshes, shipped as data files). Cases 1–4 are four plates: a rectangle with a uniform mesh (control), an L-plate with one heat source, an L-plate with two sources, and a plate with a hole under uniform tension (a stress-analysis mesh). Cases 5–9 form a complexity ladder on one L-plate, each adding one ingredient: three heat sources (refinement), a heavy-material zone (3× work per triangle), a strong-coupling zone (3× border cost) and a priority zone that should go to the fast device. Mesh density follows the features, so the work per region and the border costs are uneven. Zones and factors were fixed before any quantum result.

Table 1: Results at n=12 qubits (nine cases, noiseless). Training circuits are means over cases and scenarios; total evaluations = (training + 1) × 1 000 for quantum methods, candidates for classical ones (spectral bisection: $2(n-1)$ threshold cuts). Per shot: probability of cost ≤ 1.05× optimum (%); hit@1000: probability that 1 000 shots/candidates contain such a split (%). †Starts from the classical spectral split.

				equal devices (α=0.5)		CPU+GPU (α=0.8)	
Method	2q gates	training circuits	total evaluations	per shot	hit@1000	per shot	hit@1000
Uniform random sampling	0	0	1 000	0.07	48	0.04	29
Random with target share	0	0	1 000	0.07	48	0.41	74
Simulated annealing	0	0	1 000	–	64	–	87
Local search (restarts)	0	0	1 000	–	84	–	100
Spectral bisection	0	0	22	–	67	–	100
LR-QAOA, $p = 2$	132	0	1 000	0.27	85	0.78	99
QAOA + COBYLA, $p = 2$	132	196	196 556	0.36	94	1.1	100
CVaR-QAOA + COBYLA, $p = 2$	132	31	31 778	0.31	90	0.87	99
Warm start (spectral)†, $p = 2$	132	200	201 000	7.8	94	21	100
varQITE ZY (paper, all pairs)	66	8 675	8 675 556	78	100	<0.01	<0.01
varQITE A (all pairs)	66	7 566	7 567 222	24	33	<0.01	0.01
varQITE A^+ (edges, our ext.)	22	5 003	5 004 222	<0.01	<0.01	<0.01	<0.01
varQITE B (IonQ-native, edges)	22	4 686	4 687 167	0.01	13	0.01	7.8

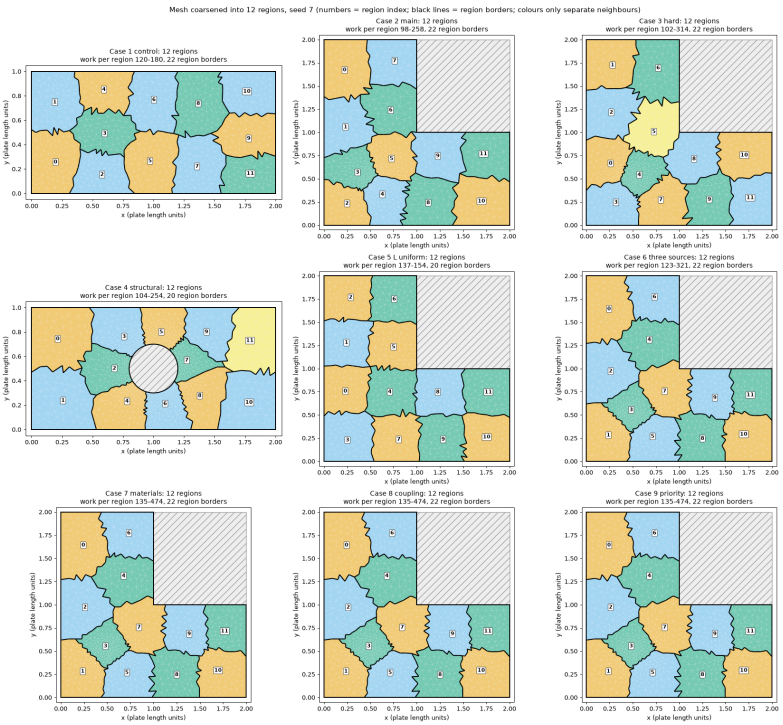


Figure 2: The nine benchmark meshes, each grouped into 12 regions (one qubit per region); numbers are region indices, black lines region borders.

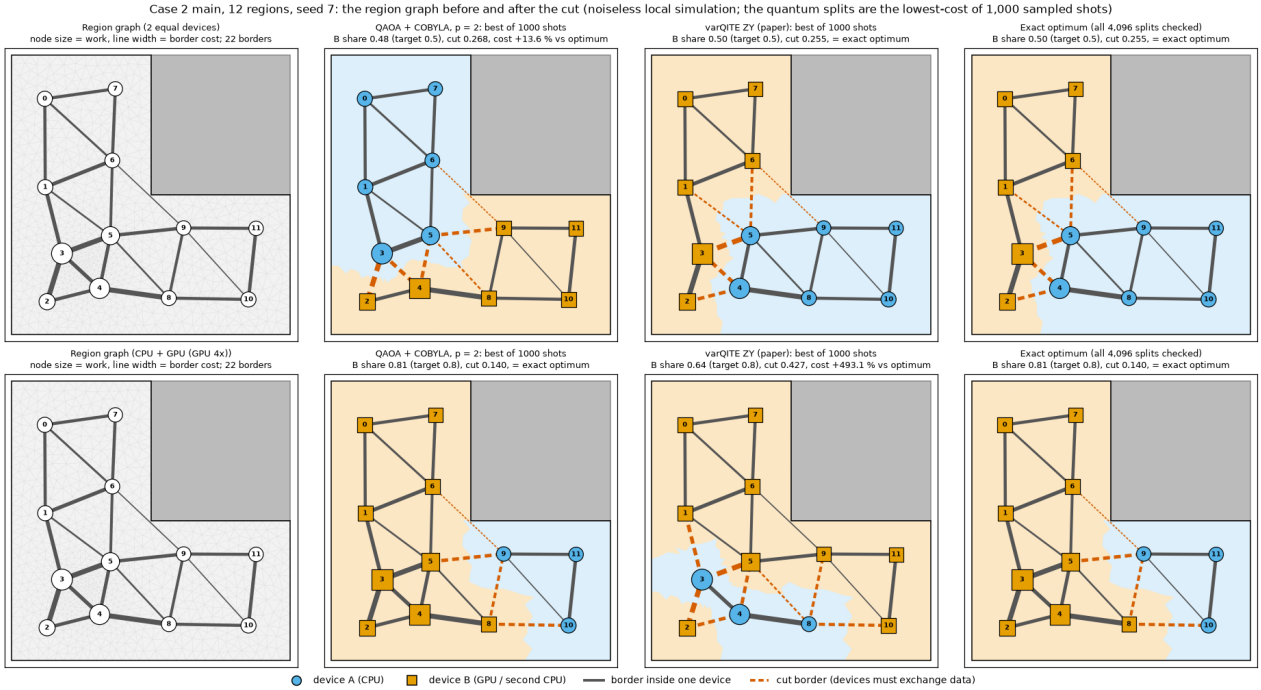


Figure 3: Case 2 (L-plate with a heat source), 12 regions. Columns: the region graph (node size = work, line width = border cost); the split returned by QAOA+COBYLA (p=2, best of 1 000 shots); the split returned by varQITE ZY; the exact optimum. Rows: equal devices (top), CPU+GPU with α=0.8 (bottom). varQITE finds the optimum for equal devices but not for the unequal share; QAOA's best shot is optimal for α=0.8.

tions count everything a method spends, in hardware-equivalent cost evaluations: (training circuits$+1)\times 1\,000$ shots for quantum methods and the number of candidates for classical ones. **Honesty measures**: schedules, step sizes and annealing temperatures are tuned on separate calibration meshes; the p=0 control and the target-share sampler are always reported; a crude depolarising estimate $F{=}(1{-}0.005)^{N_{2q}}$ indicates noise sensitivity.

4 The control-qubit limit of the varQITE ansatz

Lemma 1. *Let $U(\theta)$ be a product of gates that each commute with Z_r for a fixed qubit r (single-qubit gates on other qubits, CX with control r, $e^{i\theta Z_r Y_q}$, any gate diagonal on r). Then for every θ the measured marginal of r on $U(\theta)|{+}\rangle^{\otimes n}$ is exactly $1/2$, so every bitstring $x^\star$ has probability at most $1/2$.*

Proof. $P(x_r{=}1) = \langle\psi|\frac{1-Z_r}{2}|\psi\rangle$ with $\psi = U|{+}\rangle^{\otimes n}$; since $[U, Z_r] = 0$ this equals $\langle{+}|\frac{1-Z_r}{2}|{+}\rangle = \frac{1}{2}$, and $P(x^\star) \leq$

Table 2: hit@1000 (%) versus qubits n (cases 2, 9; seeds 7–9).

	equal devices				CPU+GPU			
n	12	16	20	24	12	16	20	24
Local search	86	45	18	10	87	96	77	52
Spectral bisection	50	33	50	67	83	100	100	100
Simulated annealing	65	32	16	12	82	48	39	26
Random, target share	39	5	0	0	65	17	8	9
LR-QAOA $p = 2$	76	20	1	–	99	50	12	–
QAOA+COBYLA $p = 2$	92	29	2	–	100	67	21	–

$P(x_r{=}x_r^\star) = \frac{1}{2}$. □

In the ZY ansatz ordered by vertex weight, the heaviest vertex is the Z side of every pair it touches; in ansatz A it is only ever a CX control. Both therefore contain such a qubit, which we verified numerically on all trained circuits. For α=0.5 without priority, $E(z){=}E(-z)$: the optimum comes as a pair $x^\star$, $\bar{x}^\star$, so a 50/50 qubit costs nothing, which is why the original MaxCut benchmarks did not expose it. For α=0.8 the optimum is unique and the cap binds. Adding one R_Y per qubit (A^+) breaks the commutation and removes the cap.

5 Results

12 qubits (Table 1, Fig. 3). All QAOA variants beat random sampling on every case: LR-QAOA raises the per-shot success 3.9× at α=0.5; at α=0.8 it is 22× uniform guessing but only 1.9× the target-share sampler (QAOA+COBYLA 2.6×). An ablation explains the gap: the single-region terms alone give 0.30% per shot, the pair terms alone 0.01%, both 0.78%. In hit@1000, QAOA+COBYLA (92%) edges out local search (86%) at α=0.5, but it spends 196 556 total evaluations against 1 000, and exhaustive search over all 4 096 splits is exact. Warm starts inherit most of their success from the spectral split; the convex-relaxation warm start is degenerate here ($c_i{=}\alpha$ for every region).

varQITE. At α=0.5 the ZY ansatz converges to near-optimal splits (78% per shot, 30–100% on every case) with 66 two-qubit gates, at a cost of 8 675 556 total evaluations. At α=0.8 every variant fails, as Lemma 1 predicts for A and ZY; A^+ removes the cap and reaches an approximation ratio of ≈ 0.98 but still misses the 5% window. At 16 qubits (graph-edge variants), A^+ reaches r=0.965 / 0.981 and no variant exceeds <0.01% per shot.

Scaling (Fig. 4, Table 2; 216 instances, 2 928 QAOA runs on an 8-core server). At fixed depth p=2 QAOA loses about an order of magnitude per shot for every four extra qubits. At 20 qubits QAOA+COBYLA reaches 2% / 21% (α=0.5 / 0.8), against 50% / 100% for spectral bisection and 18% / 77% for local search.

Noise and outcome. The depolarising estimate lowers LR-QAOA (p=2, α=0.8) from 0.78% to 0.42% per shot and makes deeper circuits worse. The modelled runtime of every method's best split lies within 2.9% of the runtime-optimal split, so better samplers matter little for this objective at this size.

6 29 qubits on HPC: the best settings against classical methods

Set-up. 29 qubits is the size of IonQ's Forte 1 noise-model simulator. We took the best settings found at 20 qubits and re-tested them, without any further tuning, on cases 2 and 9 split into 29 regions (Table 3). The selection rule was fixed before the run: per scenario, (i) the fixed schedule with the highest mean per-shot success at n=20 among all LR schedule families and depths $p \le 5$; (ii) QAOA+COBYLA at its best depth (p=2), transferred as the mean of its six optimised angle sets, whose spread across instances is at most 0.15 rad; (iii) LR-QAOA p=2, for continuity with Table 2. A dedicated complex64 statevector code builds the energy of every split from two 2^{15}-state halves and applies the mixer in cache-sized blocks, so one instance fits in 7 GB. It agrees with the notebook's float64 simulator to 3×10^{-9} in every probability at 12, 16 and 20 qubits. The 12 instances ran in parallel on two 32 GB cloud servers (about 6 minutes each). Because every energy is computed anyway, every method, classical ones included, is graded against the exact optimum, unlike the 24-qubit column of Table 2.

Result. The best fixed schedule puts 7.5×10^{-6} / 7.1×10^{-6} of its probability on near-optimal splits (α=0.5 / 0.8). That is 1 204× / 2 279× uniform guessing, so the settings transfer and QAOA still concentrates probability on good splits. At α=0.8, though, most of that gain is the work share itself: a biased coin that sends each region to the GPU with probability α does better per shot (QAOA reaches 0.7× its success), in line with the 12-qubit ablation where the single-region terms carried most of the effect. In absolute terms QAOA is far behind: 1 000 shots contain a near-optimal split with probability 0.74% / 0.7%. With the same 1 000 evaluations, simulated annealing succeeds in 10% / 43% of its runs and local search in 5.0% / 45%. Spectral bisection (56 evaluations) is optimal on 50% / 100% of the instances. The best of 1 000 QAOA shots is on average as far from the optimum as the annealing result at α=0.5 and much further at α=0.8 (gap columns). The noise estimate removes even the per-shot gain on hardware: $F{=}0.995^{2030}{\approx}3.8\times10^{-5}$ for p=5, so the circuit would sample essentially uniformly. varQITE was not run at this size: one training step of the ZY ansatz needs about 2×406 parameter-shifted simulations of 2^{29} amplitudes (estimated at days per step on these servers).

7 One batched IonQ job

Table 4 lists the 12 circuits prepared for a single IonQ Forte 1 noise-model job on Qollab (case 2, 12 qubits, 1 000 shots each): the p=0 control, LR-QAOA, QAOA+COBYLA and the final trained varQITE circuits. varQITE is trained on a noiseless local simulator; only its final circuits are sampled on IonQ. The Qollab program rebuilds the circuits from a data file, re-checks the exact optimum by brute force and grades every shot. The job has not been run yet.

8 Discussion and limitations

The heterogeneous share is not a cosmetic generalisation: it breaks the spin-flip symmetry that balanced benchmarks enjoy, and with it the success of a published ansatz. QAOA, whose mixer acts on every qubit, does not have

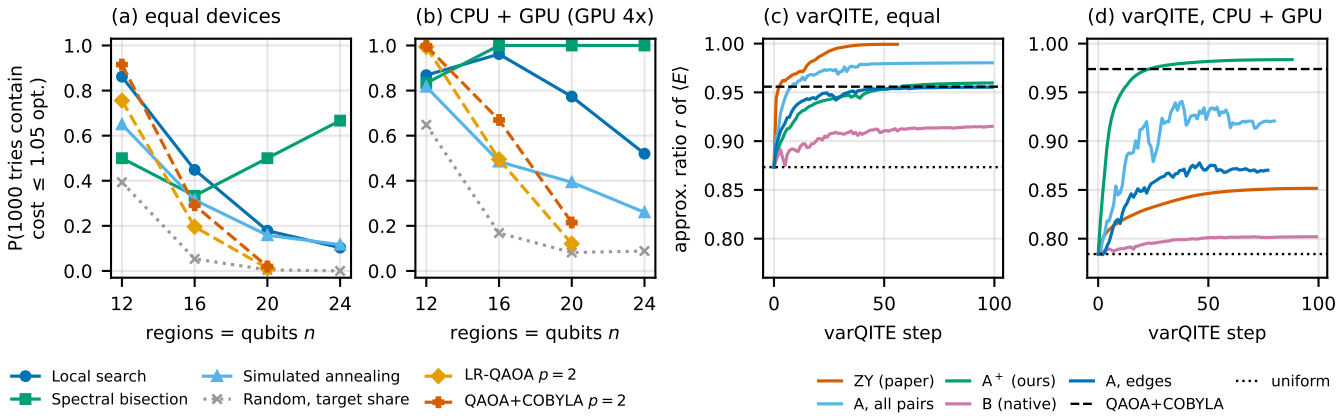


Figure 4: (a,b) Probability that 1 000 candidates (classical) or shots (QAOA, noiseless) contain a split within 5% of the optimum, versus problem size (cases 2 and 9, seeds 7–9; at 24 qubits the reference is the best classical result and QAOA is not simulated). (c,d) varQITE training curves at n=12 (mean over nine cases), with QAOA+COBYLA and uniform sampling for reference.

Table 3: 29 qubits (cases 2 and 9, seeds 7, 8, 9: 12 instances; noiseless; exact optimum by enumerating all 536 870 912 splits). Per shot: probability of cost $\leq 1.05\times$ optimum (exact for the random samplers and QAOA). hit@1000 (%): probability that 1 000 candidates or shots contain such a split, exact for the random samplers and QAOA, the share of 100 independent runs for annealing and local search, and the share of instances for the deterministic spectral bisection. Gap (%): cost of the best of 1 000 above the optimum (classical: mean over 100 runs; QAOA: one seeded 1 000-shot sample per instance, averaged over instances). The quantum settings were fixed at 20 qubits and never tuned on these instances; their training (COBYLA at n=20, LR calibration at n=12) is *not* included in the 1 000 evaluations. [a]Equal devices: LR schedule with the paper default $(\Delta\gamma, \Delta\beta)=(0.6, 0.3)$; CPU+GPU: success-calibrated LR schedule. [b]Mean of the six COBYLA-optimised angle sets at n=20.

			equal devices (α=0.5)			CPU+GPU (α=0.8)		
Method	2q gates	evaluations	per shot	hit@1000	gap	per shot	hit@1000	gap
Random, uniform	–	1 000	6.2×10^{-9}	6×10^{-4}	234	3.1×10^{-9}	3×10^{-4}	460
Random, target share	–	1 000	6.2×10^{-9}	6×10^{-4}	233	1.0×10^{-5}	1.0	135
Simulated annealing	–	1 000	–	10	76	–	43	31
Local search	–	1 000	–	5.0	72	–	45	32
Spectral bisection	–	56	–	50	13	–	100	0
Best fixed schedule[a], p=5	2 030	1 000	7.5×10^{-6}	0.74	77	7.1×10^{-6}	0.7	132
QAOA+COBYLA angles[b], p=2	812	1 000	1.9×10^{-7}	0.019	156	4.6×10^{-6}	0.46	146
LR-QAOA p=2	812	1 000	9.8×10^{-8}	0.0098	169	2.0×10^{-6}	0.2	163

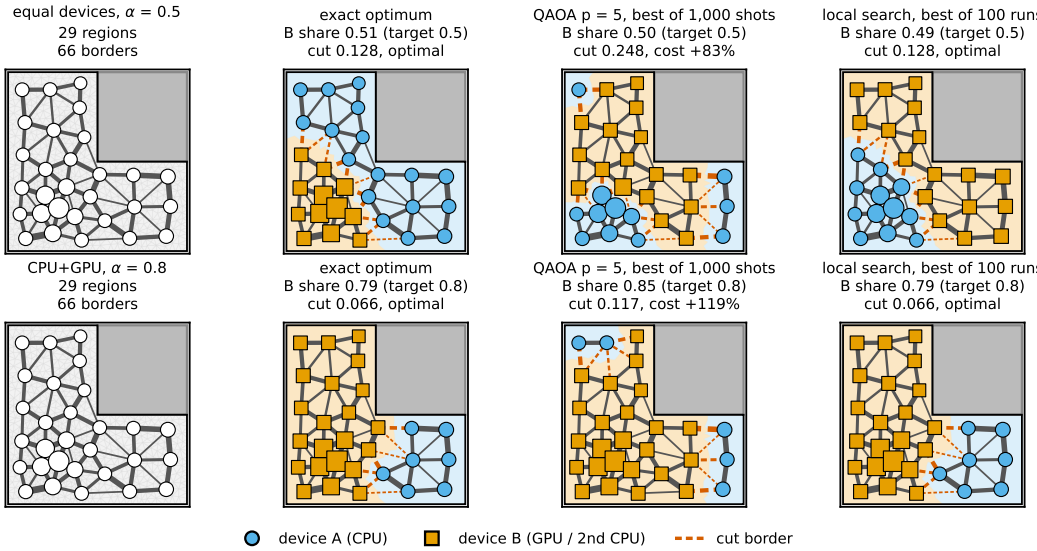


Figure 5: Case 2 at 29 qubits (seed 7). Columns: the region graph (node size = work, line width = border cost); the exact optimum among all 536 870 912 splits; the best of 1 000 noiseless QAOA shots (best fixed schedule, p=5); the best of 100 local-search runs (1 000 evaluations each). Rows: equal devices, CPU+GPU.

this blind spot but is outpaced by classical heuristics beyond 12 qubits at fixed depth; the 29-qubit run with

Table 4: The single IonQ batch: two-qubit gates and noiseless predictions (%) of P(optimum) and P(cost $\leq$ 1.05× opt.).

		equal		CPU+GPU	
Circuit	2q	opt	≤1.05	opt	≤1.05
p = 0 control (uniform)	0	0.05	0.05	0.02	0.05
LR-QAOA p = 1	66	0.13	0.13	0.31	0.60
LR-QAOA p = 2	132	0.14	0.14	0.46	0.89
QAOA + COBYLA p = 2	132	0.22	0.22	0.53	1.0
varQITE ZY (paper)	66	75	75	<0.01	<0.01
varQITE A^+ (edges)	22	<0.01	<0.01	<0.01	<0.01

the best settings found at 20 qubits confirms this gap at the largest size an IonQ simulator offers. All results are noiseless simulations or models; at 29 qubits the angles are transferred, not re-optimised; varQITE training used exact expectations; device speeds, runtimes and energies are assumptions. Next steps: run the IonQ batch; build single-qubit layers like A^+ into deeper varQITE ansätze; recursive splits for more than two devices; measured CPU/GPU runs on refined meshes.

This effort is supported by Qollab & IonQ. Code, data files and all results: `https://github.com/Oussema-t/zollerqu-qollab` (branch `mesh`). AI assistants (Claude Code, ChatGPT) were used for code and text; every number is produced by the published code.

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